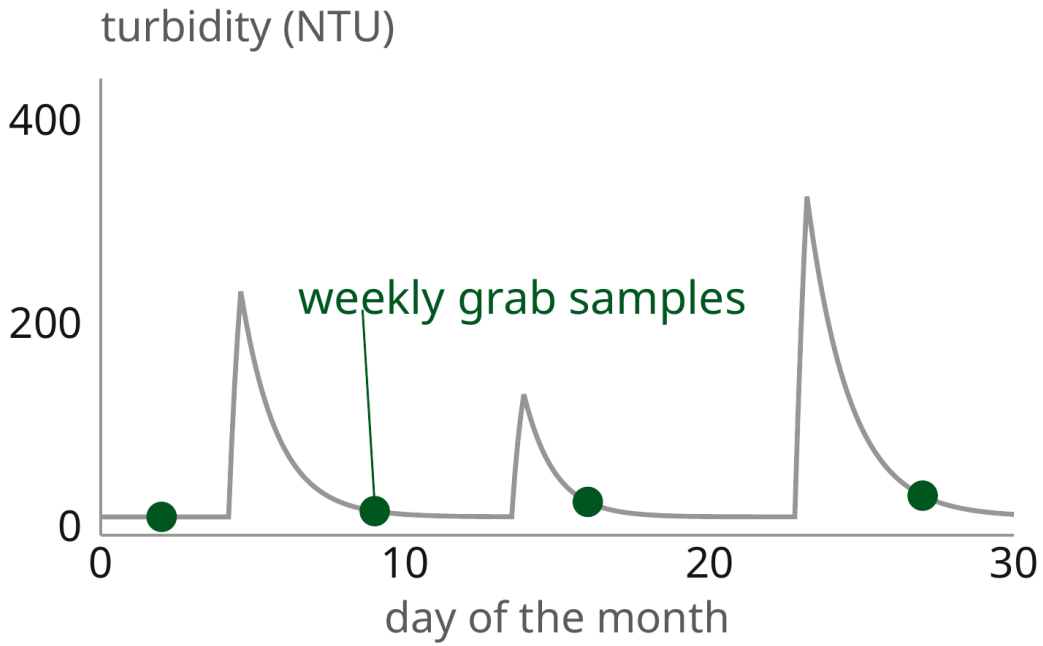
[Logo]

A low-cost optical sensor for storm-driven river turbidity

Continuous readings within 5 % of a laboratory instrument



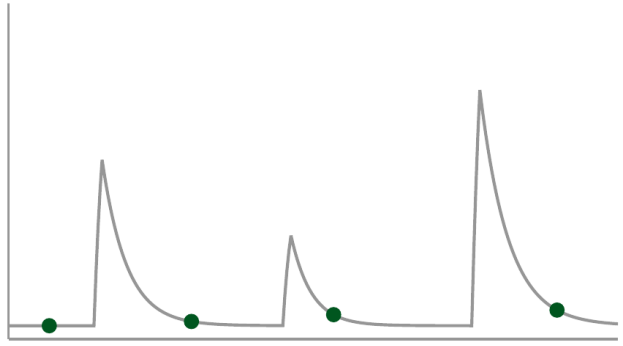
[Presenter name]

[Department] · [University]

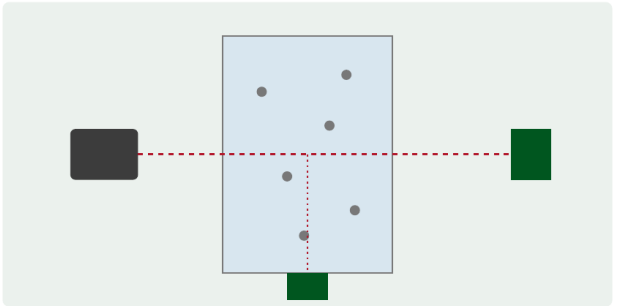
[Conference] · [Date]

[Logo]

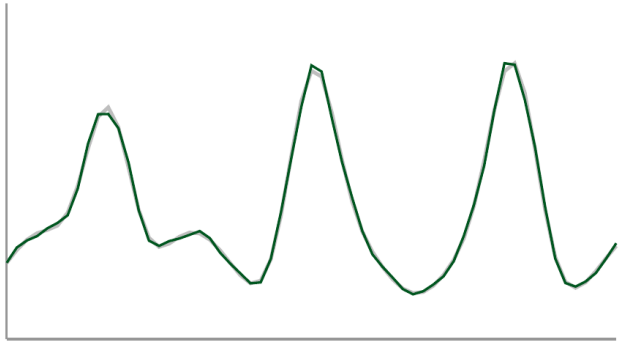
Outline



1. Need



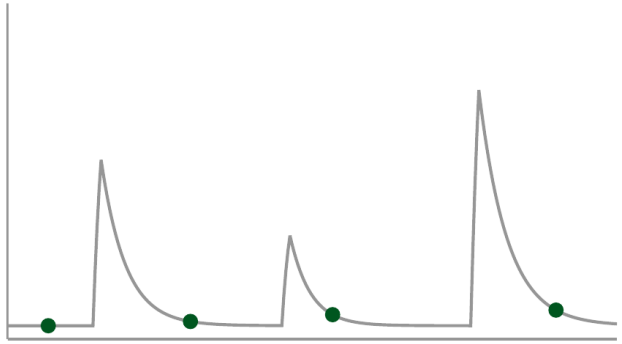
2. Sensor



3. Field test

[Logo]

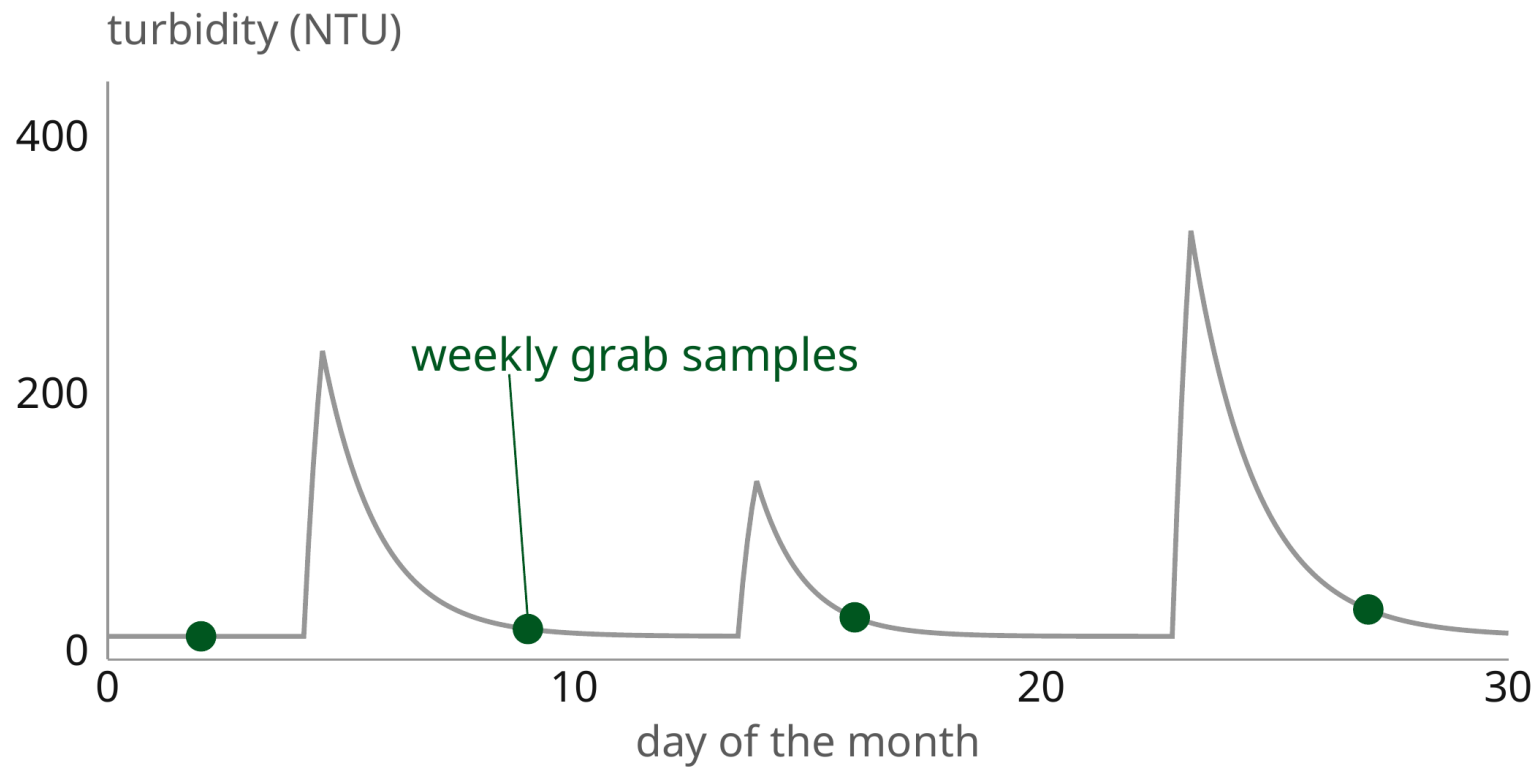
1. Need
2. Sensor
3. Field test



[Logo]

Storm loads

Storms carry most of the sediment, yet weekly samples miss them

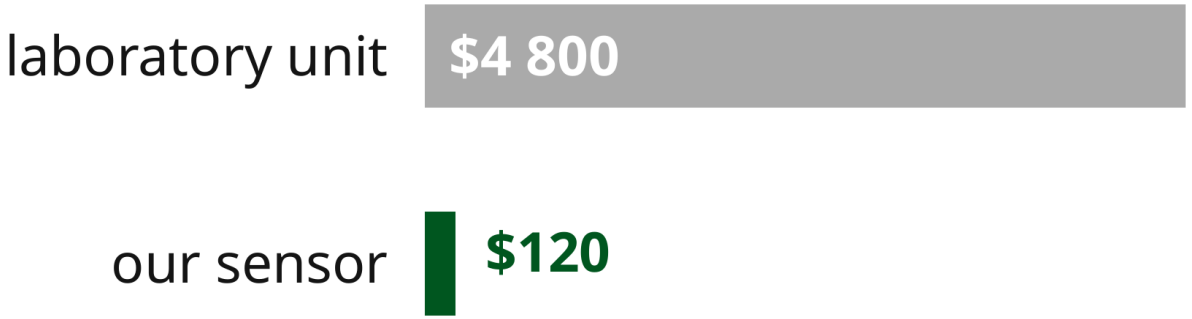


Illustrative record; sediment share after [Author, year].

[Logo]

Cost

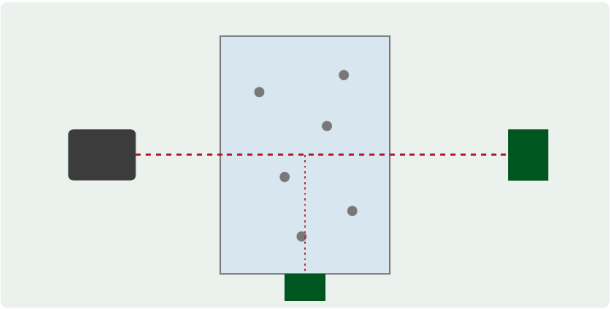
A laboratory turbidimeter costs about 40 times the parts of our sensor



Illustrative prices for one site, [year].

[Logo]

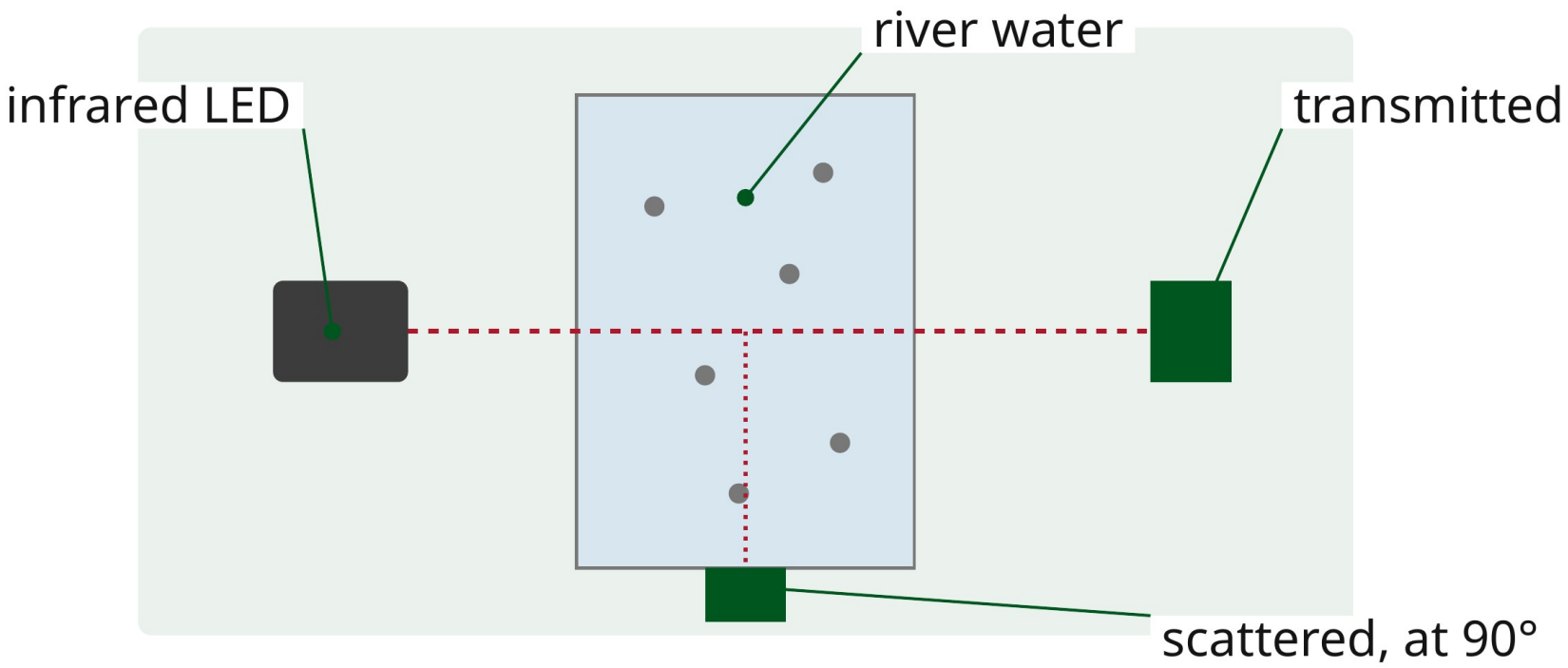
1. Need
2. Sensor
3. Field test



[Logo]

Optics

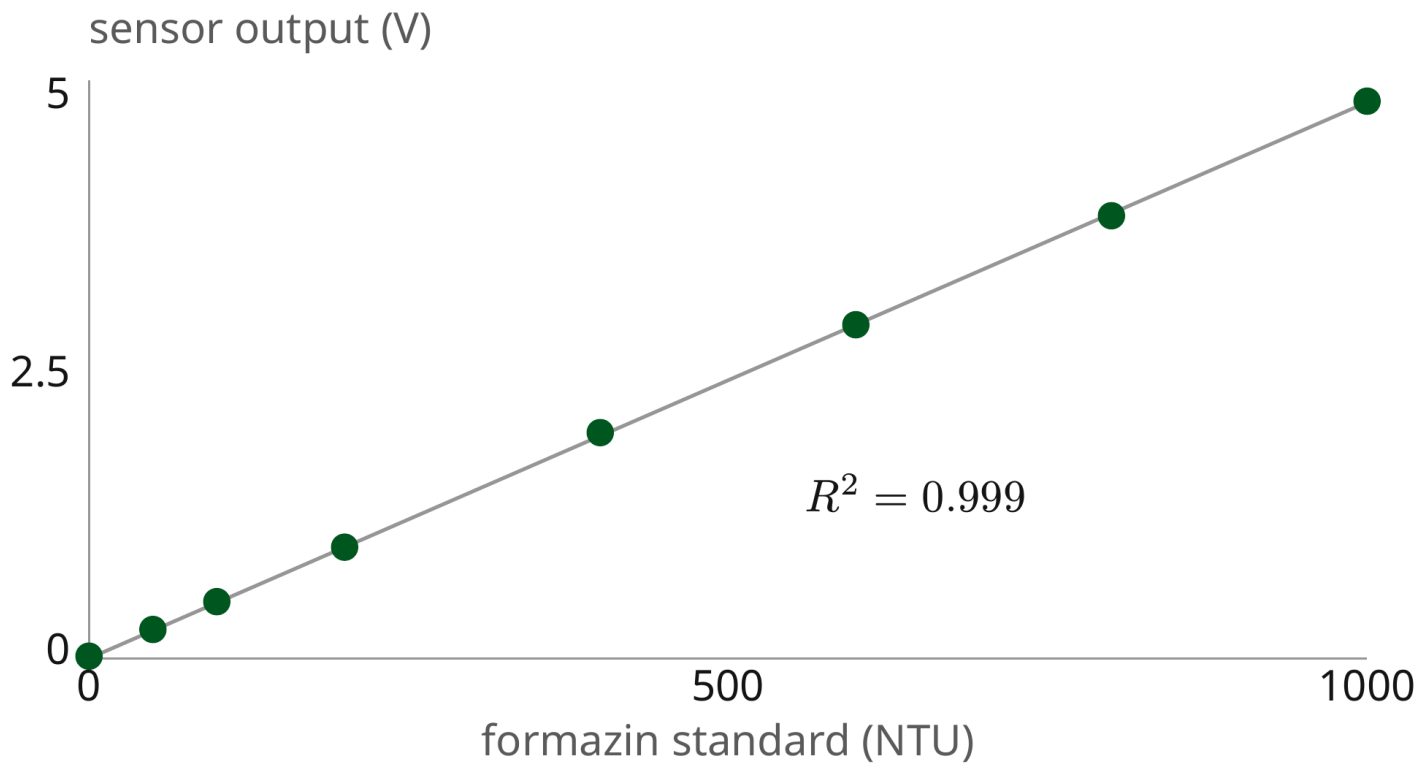
An infrared LED and two photodiodes measure scattered and transmitted light



[Logo]

Calibration

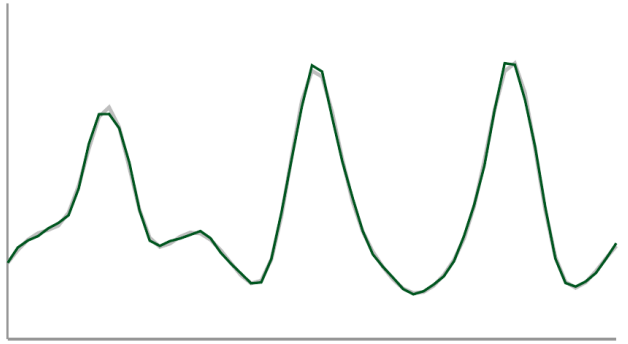
The output is linear against formazin standards up to 1000 NTU



Illustrative calibration, eight standards.

[Logo]

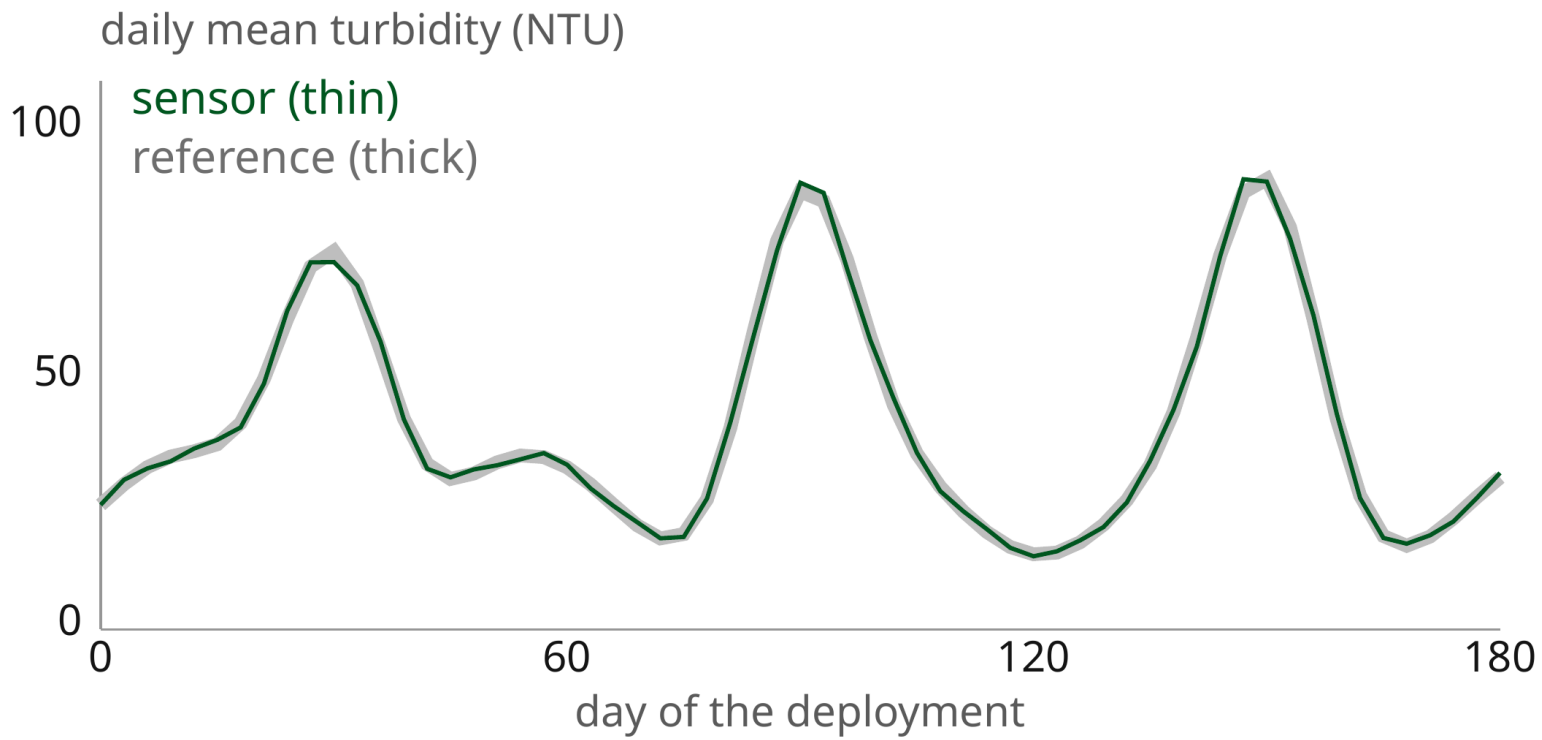
1. Need
2. Sensor
3. **Field test**



[Logo]

Agreement

Over six months, daily means stayed within 5 % of the reference

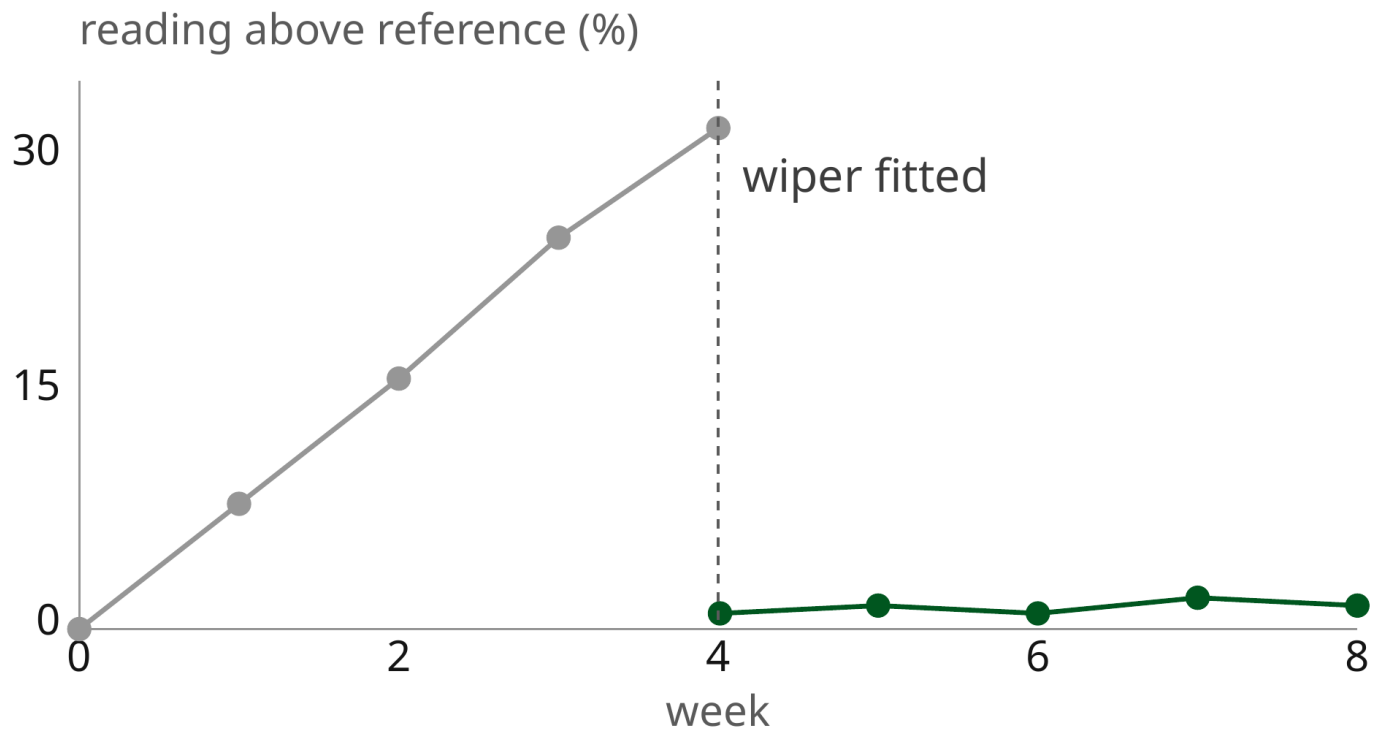


Illustrative record, [River name], [dates].

[Logo]

Fouling

Algae raised readings by 8 % a week until a wiper cleared the window

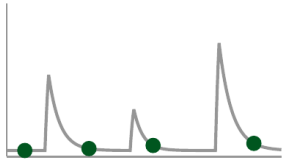


Illustrative data, first deployment.

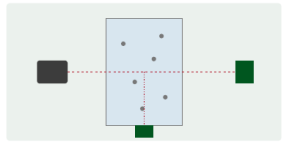
[Logo]

Conclusion

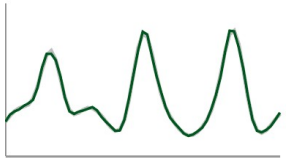
In summary, a $120 sensor records storm-driven turbidity within 5 % of a laboratory instrument



1. Weekly samples miss the storms that carry most sediment



2. Scattered and transmitted light give a linear, drift-free reading



3. With a wiper, six months stayed within 5 % of the reference

[Photo]

[Group name] · [email@example.org]

Questions?

[Logo]

Power budget

The wiper uses 2 % of the battery budget

Load	Energy per day (Wh)	Share
Optics and logger	1.90	93 %
Radio	0.10	5 %
Wiper	0.04	2 %

Illustrative values.